SN1 vs. SN2 Definition

SN1: Unimolecular, 2 steps, Rate = $k[RX]$

1. Leaving group leaves forming a carbocation
2. Nucleophile attacks C^+
(3.) (Acid-Base to balance charges)

Favored by:
- $3^\circ > 2^\circ$ (1° none)
- weak/neutral nucleophiles
- protic solvents

Partially racemic products

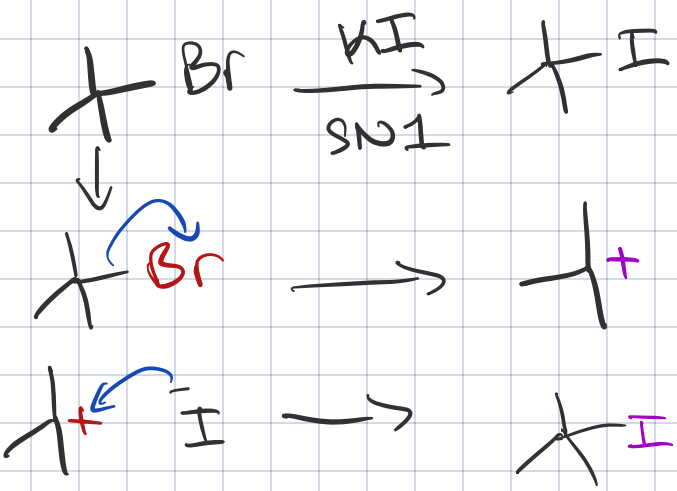
SN2 Bimolecular, 1 step, Rate = $k[RX][Nu^-]$

Favored by – $1^\circ > 2^\circ$ (3° hindered) (Me–X always!)

- aprotic solvents
- strong nucleophiles

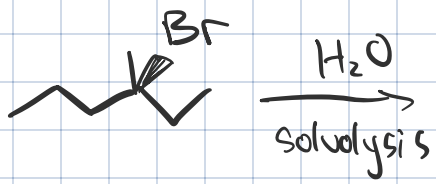
Mechanisms:

SN1:



Ex. 2 for SN1

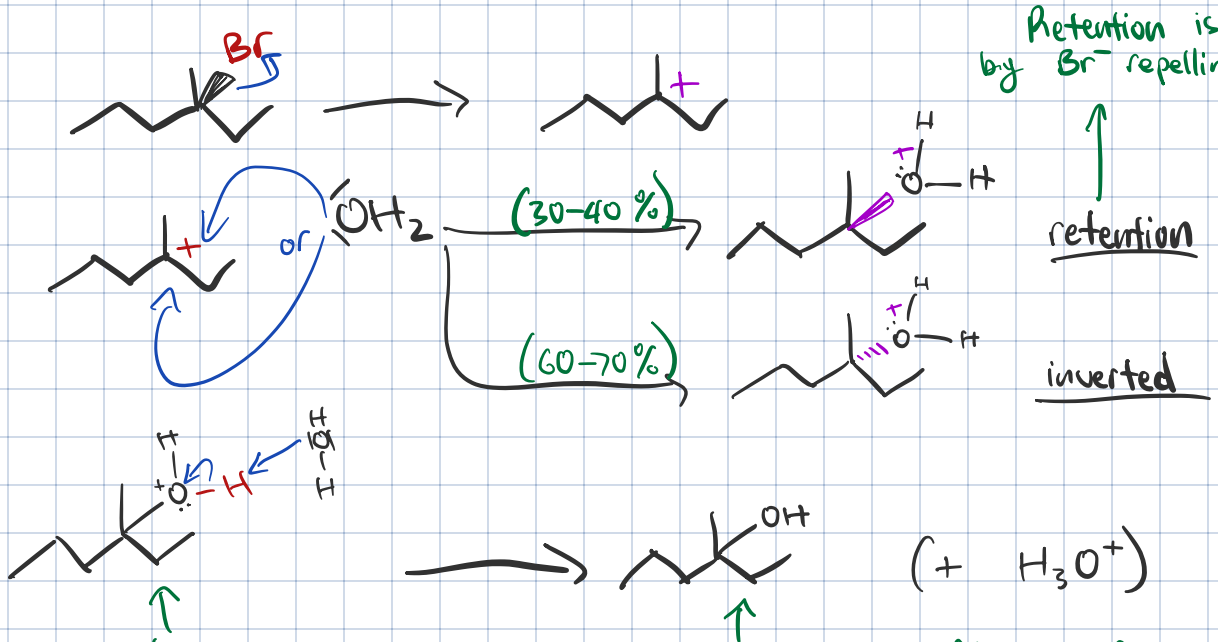
when the solvent is the nucleophile, its called solvolysis



leaving group leaves 1.

H_2O as nucleophile 2.

H_2O as base 3.

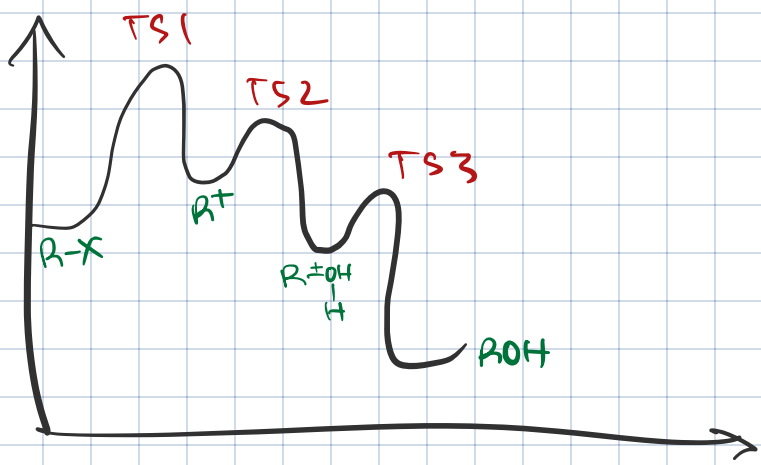


Retention is hindered by Br^- repelling $O^{\delta-}$

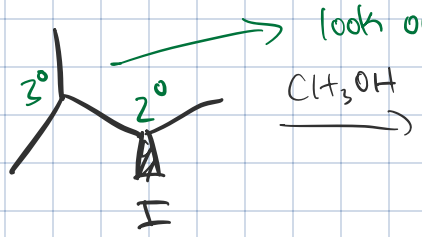
(+ H_3O^+)

w/ stereochem, which has been intentionally left out for convenience

The rxn above has 3 steps:



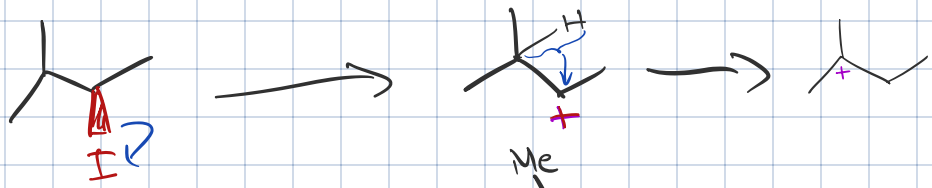
Another Ex. for SN1



look out for carbocation rearrangement (H^- shift)

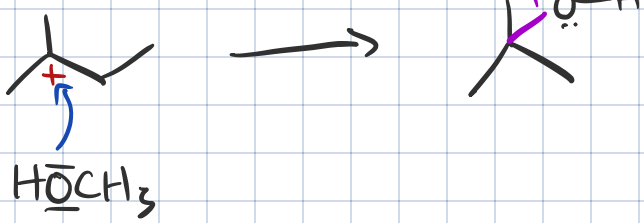
Leaving group leaves, H^- shifts

1.



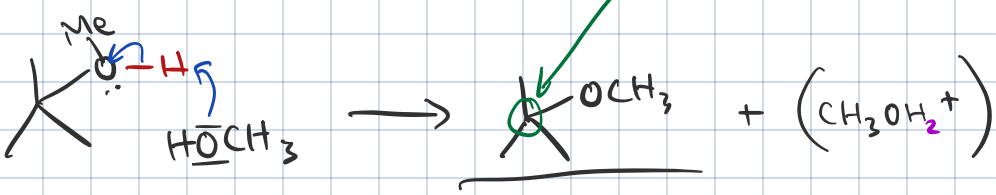
MeOH as Nuph

2.



MeOH as base

3.



achiral C, no stereochem

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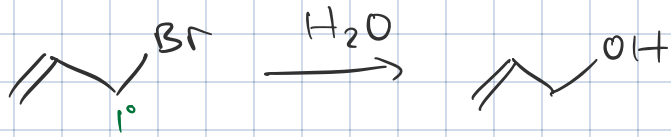
allylic primary alkyl halide (exception to 3°>2°>>1°)

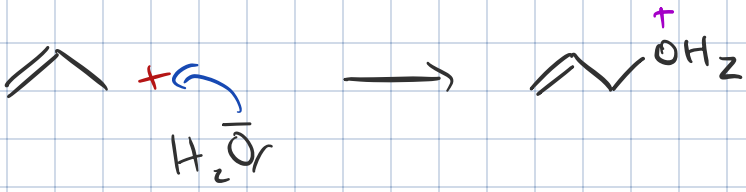


LG leaves + resonance

1.

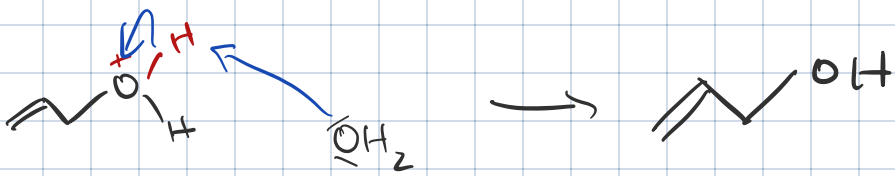
Nu

2.

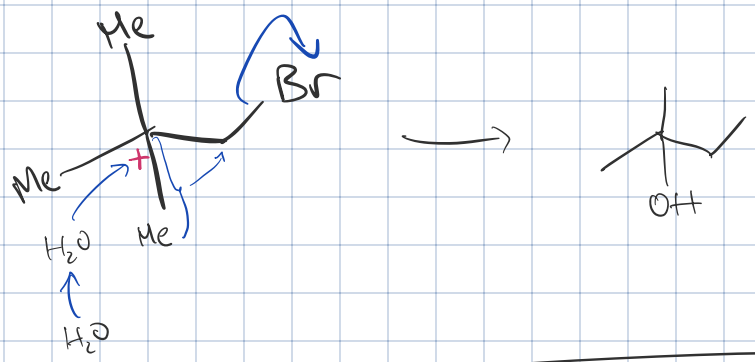


A/B

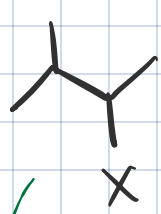
3.



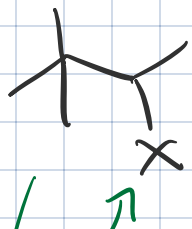
Neopentyls → methyl shift (going to combine steps from now on)



Which one favors SN1 most?

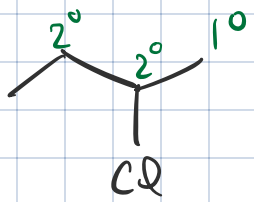


vs.



vs.

Worst



Good, competes w/ SN2

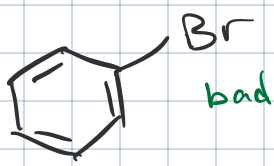
H^- shift

Me^- shift

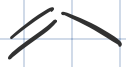
Best, because sterics make SN2 inviable

Which one favors SN1 most?

1. Step 1 breaks aromaticity
2. Aryl–X bonds are stronger than normal due to the inductive effect (hence why Halogens are activating)



bad



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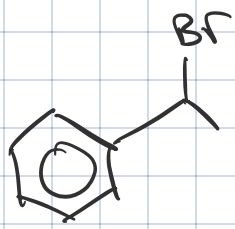
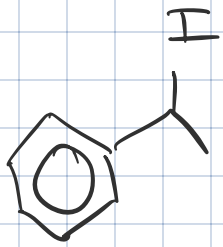


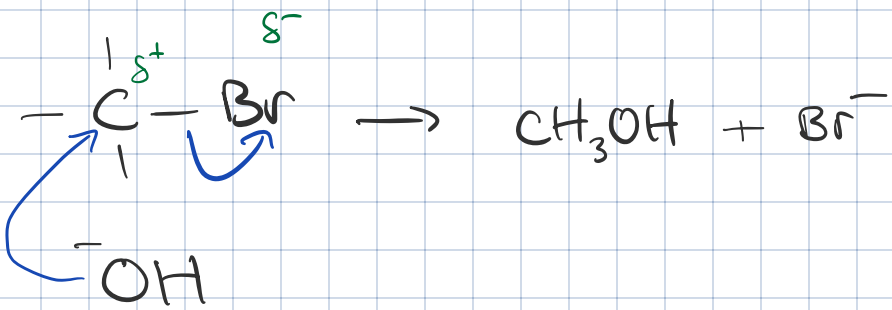
good



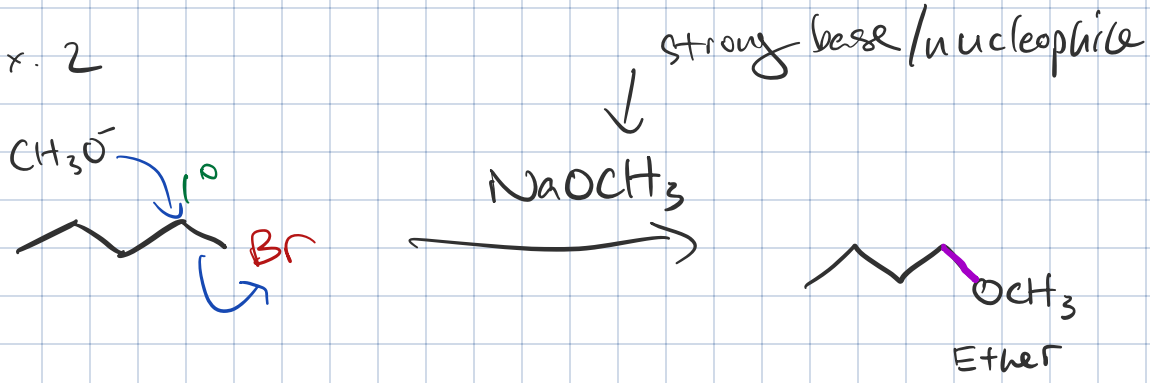
better

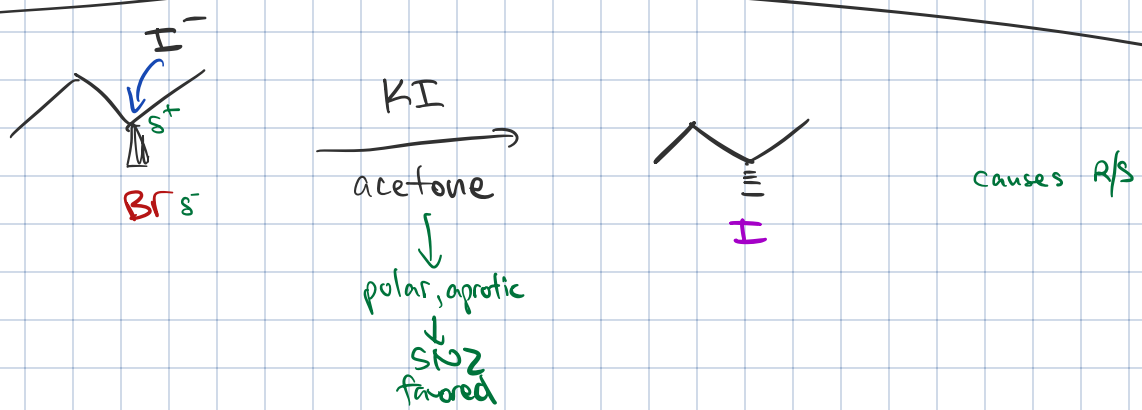
SN2 rxn mech

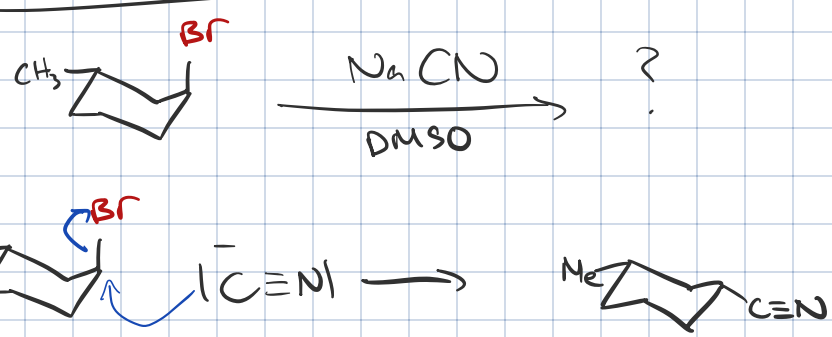
Ex. 1



Ex. 2

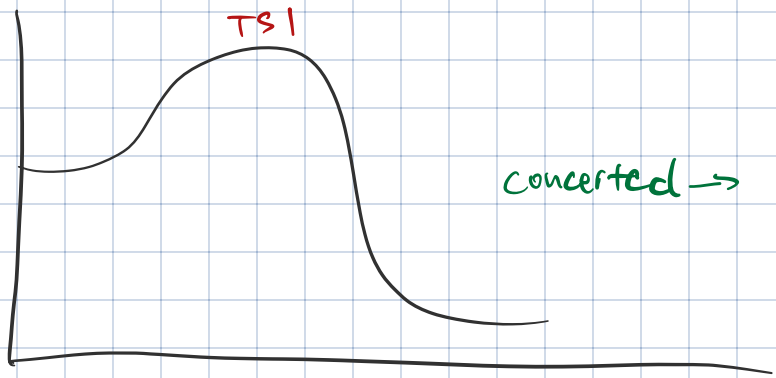




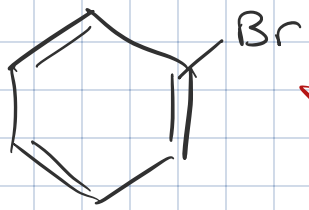


note that the substitute went from ax to eq

Rate = k[sub][Nuc]



concerted → 1 step



✗ Aryl Halides do not do SN2, π system repels nucleophiles, and transition state is too high (need to make a sp^2 aromatic carbon sp^3)

Nucleophillic Strength

Decrease →

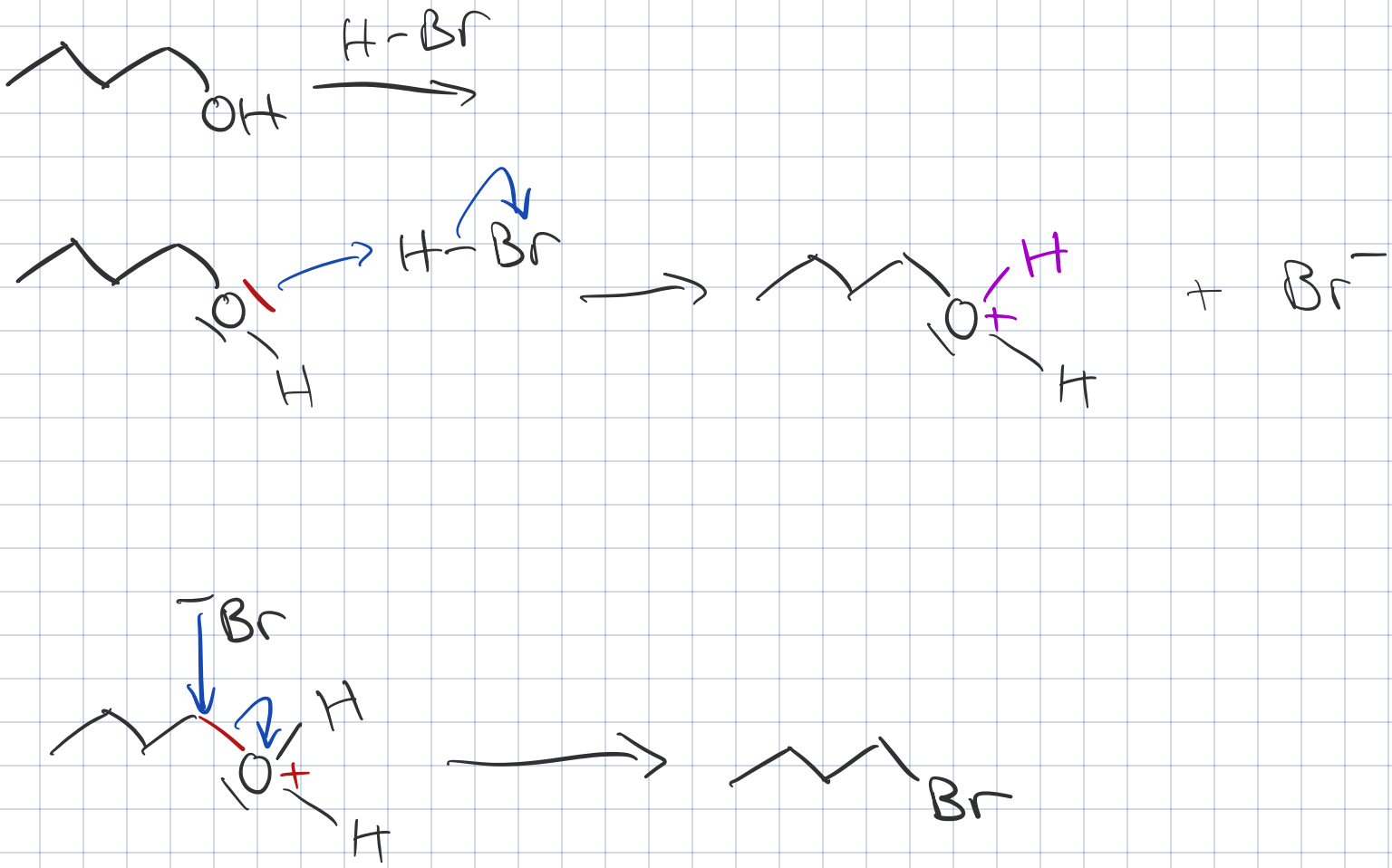
C N O F

Cl

Br

I

Decrease if protic due to stability of harder anions from H-bonds.

↳ Increase in aprotic solvents due to basicity trend

Another Ex for SN2